Boolean operators and truth tables

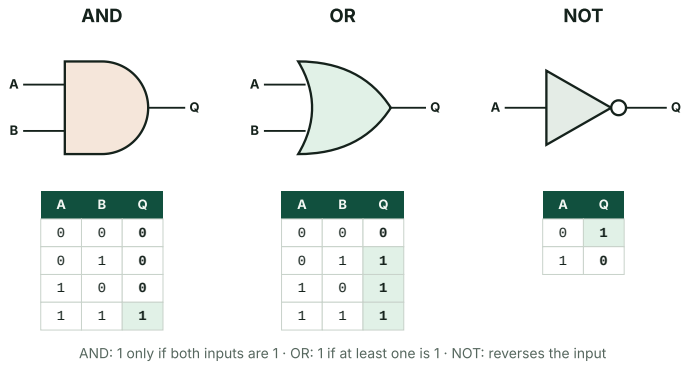
D08 FACT FILE

Specification: students should be able to

- Demonstrate understanding of and use Boolean operators (AND, OR and NOT) and truth tables

CCEA fact file: Digital Data 2

Key words: Boolean, AND, OR, NOT, truth table, logic gate, condition, relational operator



AND, OR and NOT logic gates with their truth tables

Operators

AND gives True only when both inputs are True. OR gives True when at least one input is True. NOT reverses its input. A truth table lists every possible combination of inputs and the output; with n inputs there are 2^n rows. For combined expressions, add an intermediate column for each bracket.

In programs

Conditions in if statements and loops are Boolean expressions. Relational operators (==, !=, <, >, <=, >=) produce true or false, and && (AND), || (OR) and ! (NOT) combine them, e.g. while (choice != "Y" && choice != "N"). A common error is using OR (||) where AND (&&) is needed in a validation loop.

T Key terms from the mark schemes

Truth table
Table showing the output for every combination of inputs.
Credited: U4 2022 Q1, 2024 Q4, 2025 Q1

AND / OR / NOT
AND is true only if both inputs are true; OR is true if at least one is; NOT reverses the input.
Credited: U4 2019 Q8, 2025 Q1

C C# example

Boolean operators in code

```
Console.Write("Age: ");
int age = Convert.ToInt32(Console.ReadLine());
bool hasTicket = true;
if (age >= 12 && age <= 16 && hasTicket)
{
    Console.WriteLine("Welcome to the disco");
}
if (!hasTicket)
{
    Console.WriteLine("Please buy a ticket");
}
```

Exam focus

Truth tables are on every paper: 2019 Q8(e), 2022 Q1(a)–(b), 2023 Q1(c), 2024 Q4(e), 2025 Q1(b) and Q1(d). Fill every row and check intermediate columns.

Unit 4 2025 Q1	[5]	String type, compound AND, IF output, NOT/AND table, bubble sort
Unit 4 2024 Q4	[19]	Bit, nibble, byte; denary–binary–hex conversions; binary addition and overflow; truth tables
Unit 4 2023 Q1	[10]	ASCII acronym and capacity; AND/OR truth table; Unicode vs ASCII
Unit 4 2022 Q1(a)–(c)	[4]	Truth tables for NOT (P AND Q), (P OR Q) AND R; tracing outputs
Unit 4 2019 Q8	[18]	Binary, hexadecimal and its uses, binary addition and overflow, logic gates

? Check yourself

1. Complete the truth table for A AND B for all four input combinations. [2]
2. Complete the truth table for NOT (P AND Q). [2]
3. How many rows does a truth table with three inputs need? Explain. [2]

Answers: 1. A B | A AND B: 0 0 → 0 / 0 1 → 0 / 1 0 → 0 / 1 1 → 1. Award: first three rows all 0 (1) / 1 1 → 1 (1) 2. P Q | P AND Q | NOT (P AND Q): 0 0 → 0 → 1 / 0 1 → 0 → 1 / 1 0 → 0 → 1 / 1 1 → 1 → 0. Award: 1, 1, 1 for the first three rows (1) / 0 for P = 1, Q = 1 (1) 3. 8 rows (1) / because each input can be 0 or 1, so there are $2 \times 2 \times 2 = 2^3 = 8$ combinations (1)